

3 (a) Define power.

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 [1]

(b) An electric car is powered by a motor. The car is travelling at a constant speed of 35 ms^{-1} along a straight horizontal road, as shown in Fig. 3.1.

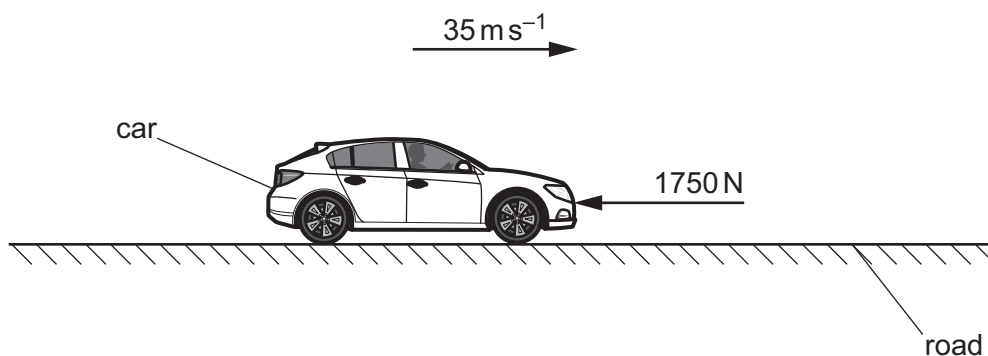


Fig. 3.1

There is a total resistive force of 1750 N acting on the car.

(i) Calculate the power transmitted to the wheels of the car by the motor.

power =W [2]

(ii) Calculate the useful work done by the motor when the car travels a distance of 17 km .

work done =J [2]



- (iii) The potential difference (p.d.) across the motor has a constant value of 600 V and the motor has an efficiency of 85%.

Calculate the current in the motor.

current = A [3]

- (c) The car in (b) now reaches a slope, as shown in Fig. 3.2.

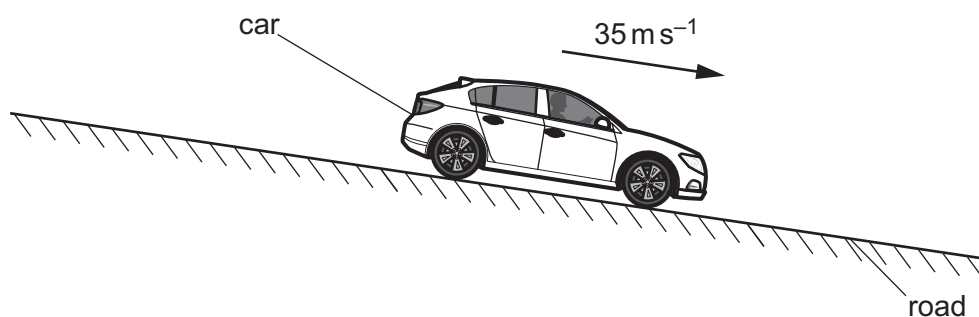


Fig. 3.2

The car continues down the slope at the same speed as in (b).

State and explain the effect, if any, of the slope on:

- (i) the air resistance acting on the car

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 [1]

- (ii) the current in the motor.

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 [1]

[Total: 10]

